

Multiplicative closure of Levitan and Poisson multipliers

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Source and answer

Chaouchi, Kostić, and Velinov [1] ask whether, for $c \in \mathbb{C} \setminus \{0, 1\}$, there can be a bounded continuous $F : \mathbb{R}^n \rightarrow \mathbb{C}$ which is Levitan (N, c) -almost periodic but not Levitan (N, c^2) -almost periodic, or which is uniformly c -Poisson stable but not uniformly c^2 -Poisson stable.

2.1. Levitan (N, c) -almost periodic functions and uniformly Poisson c -stable functions. In this subsection, we will provide some examples and open questions concerning the Levitan (N, c) -almost periodic functions and the uniformly Poisson c -stable functions, where $c \in \mathbb{C} \setminus \{0\}$; for simplicity, we will always consider here the case in which $\Lambda = \mathbb{R}^n$, $\phi(x) \equiv x$, $\mathbb{F}_{\mathbf{K}} \equiv 1$ and $P_K = C(K)$ for each non-empty compact set $K \subseteq \mathbb{R}^n$. Let $\rho = cI$; then any Levitan $(\phi, \mathbb{F}_{\mathbf{K}}, \mathcal{B}, \Lambda', \rho, \mathcal{P}_{\mathbf{K}})$ -almost periodic function is simply called Levitan (N, Λ', c) -almost periodic and any Bebutov $(\phi, \mathbb{F}_{\mathbf{K}}, \mathcal{B}, \Lambda', \rho, \mathcal{P}_{\mathbf{K}})$ -uniformly recurrent function is simply called uniformly Poisson (Λ', c) -stable [we omit the term “ Λ' ” from the notation if $\Lambda' = \mathbb{R}^n$].

The proofs of [33] Proposition 4.2.14, Proposition 7.1.13] do not work for the Levitan/Bebutov concepts; because of that, we would like to ask the following:

Problem. Suppose that $c \in \mathbb{C} \setminus \{0, 1\}$. Can we find a bounded continuous function $F : \mathbb{R}^n \rightarrow \mathbb{C}$ which is Levitan (N, c) -almost periodic (uniformly c -Poisson stable) but not Levitan (N, c^2) -almost periodic (uniformly c^2 -Poisson stable)?

The following important counterexample shows that there exists a non-trivial uniformly Poisson c -stable function $F : \mathbb{R} \rightarrow \mathbb{R}$ for any complex number $c \neq 0$;

The answer is no in both cases. More generally, the multiplier classes obey a multiplicative composition law. Consequently, membership for c automatically implies membership for every positive integral power c^m . Neither boundedness nor continuity is used in this implication.

Definitions in the source’s specialization

For a compact set $K \subset \mathbb{R}^n$, a function $F : \mathbb{R}^n \rightarrow X$ into a normed complex vector space, a multiplier $a \in \mathbb{C}$, and $\epsilon > 0$, put

$$\mathcal{E}_a(K, \epsilon) := \left\{ \tau \in \mathbb{R}^n : \sup_{t \in K} \|F(t + \tau) - aF(t)\| \leq \epsilon \right\}.$$

In the specialization used immediately before the source problem, F is Levitan (N, a) -almost periodic precisely when $\mathcal{E}_a(K, \epsilon)$ is relatively dense in $\mathbb{R}^n$ for every compact K and every $\epsilon > 0$. It is uniformly a -Poisson stable precisely when, for every compact K , there is a sequence (τ_j) with $|\tau_j| \rightarrow \infty$ and

$$\sup_{t \in K} \|F(t + \tau_j) - aF(t)\| \longrightarrow 0.$$

The sequence is allowed to depend on K , exactly as in Definition 2.3(ii) of the source.

Proof intuition

Two approximate multiplier-return translations can be performed in succession. First move by an a -return on K ; then move by a b -return on the translated compact set. The composite translation returns F to abF , with the two errors joined by one triangle inequality. Relative density survives the fixed first translation. For Poisson stability, the second return can be chosen much larger than the first, preventing cancellation and forcing the composite translations to escape to infinity.

The composition theorem

Theorem 1 (Multiplicative closure). *Let $a, b \in \mathbb{C}$ and $F : \mathbb{R}^n \rightarrow X$.*

1. *If F is Levitan (N, a) -almost periodic and Levitan (N, b) -almost periodic, then it is Levitan (N, ab) -almost periodic.*
2. *If F is uniformly a -Poisson stable and uniformly b -Poisson stable, then it is uniformly ab -Poisson stable.*

Proof. Fix a compact $K \subset \mathbb{R}^n$ and $\epsilon > 0$, and set

$$\delta := \frac{\epsilon}{1 + |b|}.$$

For the Levitan assertion, choose any $\sigma \in \mathcal{E}_a(K, \delta)$. The translated set $K + \sigma$ is compact, so $\mathcal{E}_b(K + \sigma, \delta)$ is relatively dense. If $\tau \in \mathcal{E}_b(K + \sigma, \delta)$ and $t \in K$, then

$$\begin{aligned} \|F(t + \sigma + \tau) - abF(t)\| &\leq \|F((t + \sigma) + \tau) - bF(t + \sigma)\| \\ &\quad + |b| \|F(t + \sigma) - aF(t)\| \\ &\leq (1 + |b|)\delta = \epsilon. \end{aligned}$$

Therefore

$$\sigma + \mathcal{E}_b(K + \sigma, \delta) \subseteq \mathcal{E}_{ab}(K, \epsilon).$$

The set on the left is a translate of a relatively dense set, hence is relatively dense. This proves the first assertion.

For the Poisson assertion, choose from a -Poisson stability on K a sequence (σ_j) such that

$$\sup_{t \in K} \|F(t + \sigma_j) - aF(t)\| \leq \frac{1}{j(1 + |b|)}.$$

(Passing to a tail or subsequence gives this normalization.) For the compact set $K + \sigma_j$, b -Poisson stability supplies a sequence of translations tending to infinity and whose error there tends to zero. Choose one term τ_j so far out that

$$|\tau_j| > j + |\sigma_j|, \quad \sup_{s \in K + \sigma_j} \|F(s + \tau_j) - bF(s)\| \leq \frac{1}{j(1 + |b|)}.$$

Set $\rho_j = \sigma_j + \tau_j$. Then $|\rho_j| \geq |\tau_j| - |\sigma_j| > j$, while the same two-step estimate gives

$$\sup_{t \in K} \|F(t + \rho_j) - abF(t)\| \leq \frac{1}{j}.$$

Thus (ρ_j) witnesses uniform ab -Poisson stability on K . □

Corollary 1 (Power closure and resolution of the problem). *If F is Levitan (N, c) -almost periodic, then it is Levitan (N, c^m) -almost periodic for every integer $m \geq 1$. If F is uniformly c -Poisson stable, then it is uniformly c^m -Poisson stable for every integer $m \geq 1$. Hence neither function requested in the source problem exists.*

Proof. The case $m = 1$ is the hypothesis. If the conclusion holds for m , apply Theorem 1 with $a = c^m$ and $b = c$. Induction proves the claim, and $m = 2$ answers both alternatives in the problem. $\square$

Checks, scope, and novelty

The proof was checked directly against Definition 2.3 and the specialization in Section 2.1 of the source. The only structural facts used are that a translate of a relatively dense subset of $\mathbb{R}^n$ remains relatively dense, that $K + \sigma$ is compact, and the triangle inequality. In particular, the diagonal choice in the Poisson proof is necessary because the source's recurrence sequence may depend on the compact set.

The theorem applies in all dimensions and, unchanged, to functions with values in any complex normed space. It does not require boundedness or continuity. It answers exactly the displayed c versus c^2 problem; it does not address the paper's other open questions.

On 2026-08-12, the four cheap run indexes, exact-problem and exact-title searches, citation searches, and targeted searches for Levitan (N, c^2) and uniformly c^2 -Poisson stability found no later claimed resolution. This is a bounded novelty check, not an exhaustive survey of all non-arXiv literature.

Human review recommendation

Review as a likely-valid full negative solution. The highest-value checks are that the simplified definitions above match the source specialization, and that the second Poisson translation is chosen on the shifted compact set $K + \sigma_j$. Once those points are fixed, both conclusions follow from the displayed two-step triangle inequality.

References

- [1] B. Chaouchi, M. Kostić, and D. Velinov, *Metrical almost periodicity: Levitan and Bebutov concepts*, arXiv:2209.13576 (2022), especially Definition 2.3 and Section 2.1.